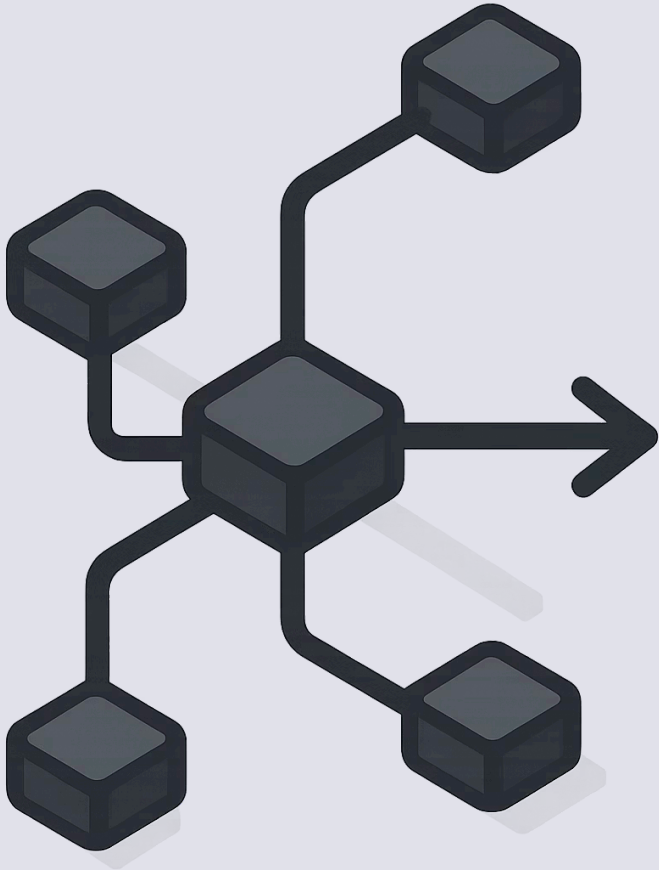




Introduction to ML Frameworks

TensorFlow, PyTorch & Scikit-learn

A beginner-friendly guide to understanding how machine learning frameworks work — with simple explanations, real examples, and Python code.

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What is an ML Framework?

An ML framework is a software library that provides ready-made tools for building machine learning models – so we don't have to write complex math from scratch.

Scikit-learn

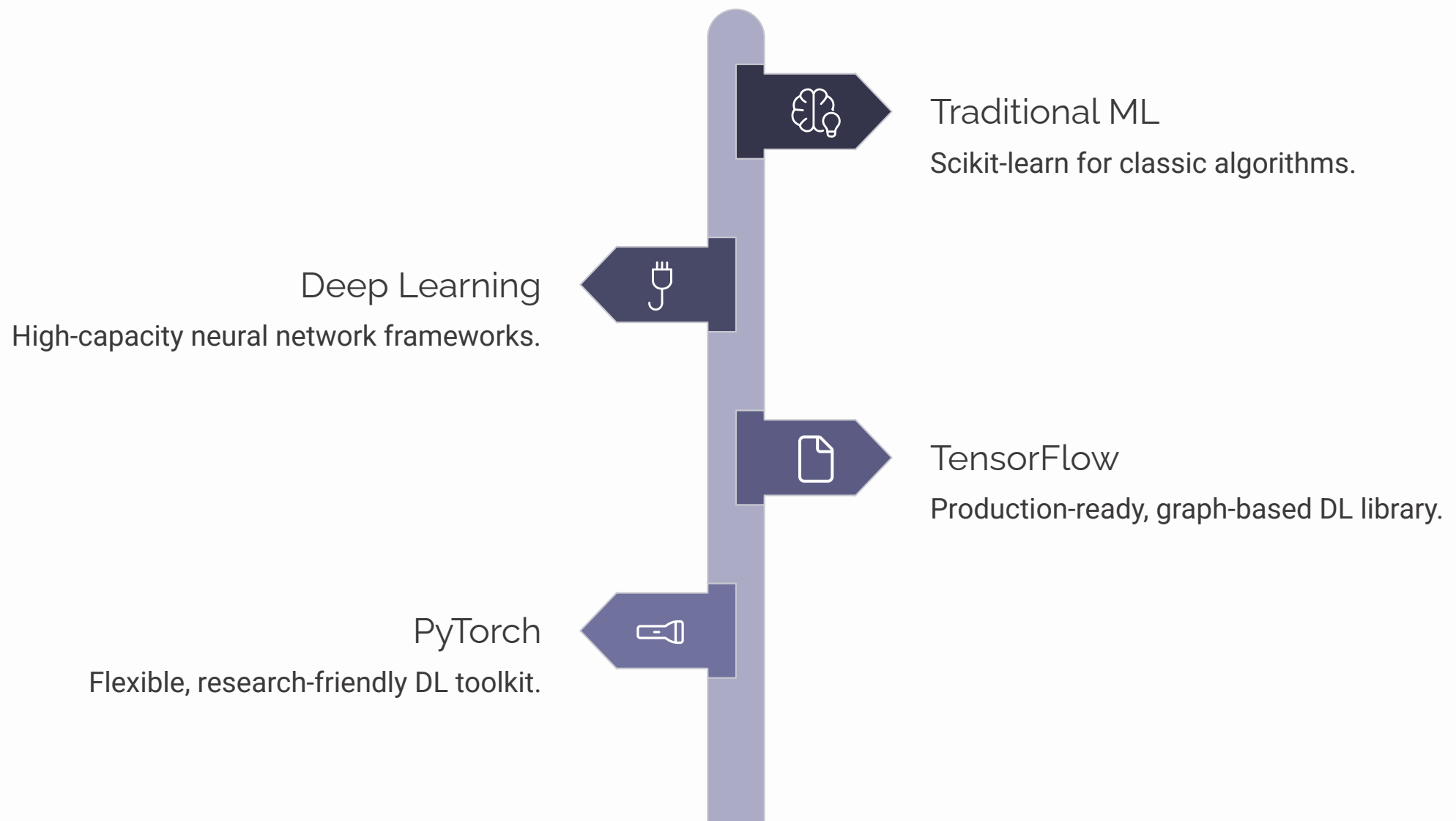
Traditional ML algorithms: Linear Regression, Decision Tree, SVM, KNN, Random Forest

TensorFlow

ML + Deep Learning: Neural Networks, CNN, RNN, Computer Vision, NLP

PyTorch

Deep Learning + AI Research: Neural Networks, Transformers, Generative AI, LLMs



Scikit-learn — Traditional ML with Python

What Scikit-learn Provides

- **Algorithms:** Linear Regression, Decision Tree, Random Forest, SVM, KNN, K-Means
- **Tools:** Data preprocessing, train-test split, cross-validation, hyperparameter tuning, model evaluation

Example: Predict Student Pass/Fail

Features used: Study Hours, Attendance, Previous Marks → Model predicts PASS (1) or FAIL (0)

```
# Import tools
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

# Features: [StudyHrs, Attendance, PrevMarks]
X = [[2,60,45],[4,75,55],[6,85,70],
      [1,50,35],[5,80,65],[7,90,80]]

# Labels: 0=Fail, 1=Pass
y = [0, 1, 1, 0, 1, 1]

# Split data into train and test sets
X_train, X_test, y_train, y_test = \
    train_test_split(X, y, test_size=0.2)

# Create and train the model
model = DecisionTreeClassifier()
model.fit(X_train, y_train) # Learning step

# Predict and evaluate
y_pred = model.predict(X_test)
print("Accuracy:", accuracy_score(y_test, y_pred))
```

TensorFlow & PyTorch — Deep Learning

TensorFlow / Keras — Neural Network

```
import tensorflow as tf
from tensorflow import keras

# Build neural network layers
model = keras.Sequential([
    # 16 neurons, relu activation
    keras.layers.Dense(16, activation="relu",
                        input_shape=(3,)),
    keras.layers.Dense(8, activation="relu"),
    # Output: 1 neuron, sigmoid for 0/1
    keras.layers.Dense(1, activation="sigmoid")
])

# Configure: optimizer, loss, metric
model.compile(optimizer="adam",
              loss="binary_crossentropy",
              metrics=["accuracy"])

# Train for 20 epochs
model.fit(X_train, y_train, epochs=20)
loss, acc = model.evaluate(X_test, y_test)
print("Accuracy:", acc)
```

PyTorch — Explicit Training Loop

```
import torch.nn as nn, torch.optim as optim

# Build model: 3 inputs → 16 → 8 → 1 output
model = nn.Sequential(
    nn.Linear(3, 16), nn.ReLU(),
    nn.Linear(16, 8), nn.ReLU(),
    nn.Linear(8, 1), nn.Sigmoid()
)

criterion = nn.BCELoss() # Loss function
optimizer = optim.Adam(model.parameters(),
                        lr=0.001)

for epoch in range(20):
    output = model(X_train) # Forward pass
    loss = criterion(output, y_train) # Error
    optimizer.zero_grad() # Clear gradients
    loss.backward() # Backpropagation
    optimizer.step() # Update weights
    print(f"Epoch {epoch+1} Loss:{loss.item():.4f}")
```

- ❏ PyTorch exposes the full training loop — making it ideal for understanding what actually happens inside neural network training.



Which Framework Should You Use?



Use Scikit-learn

Structured / tabular data with traditional ML. Examples: house-price prediction, student performance, loan classification, customer churn.



Use TensorFlow

Large datasets needing deep learning. Examples: image classification, computer vision, speech systems, deep neural networks.



Use PyTorch

Research and flexible architectures. Examples: CNN, NLP, Transformers, Generative AI, LLM research.

Key Takeaway: Frameworks change the programming interface, but the fundamental ML concept — make predictions, calculate error, update parameters, repeat — always remains the same.